FieldSpec – Product Requirements Document

Executive Summary

FieldSpec is a specialised online platform designed to bridge the gap between raw aerial imagery and actionable agricultural insights. It leverages advanced computer vision and artificial intelligence (AI) to transform drone-collected survey images into detailed, professional reports for farmers, agronomists, and land managers.

Problem Statement

Agricultural and land survey drone operators are great at flying and end up with hundreds of photos, but turning those into professional client-ready reports is a manual, time-consuming process.
Operators usually need to sort, annotate, write, and format professionally manually. Key problems include:

1. **No Unified Workflow for Photo Management and Report Building:** Operators have no single place to sort, label, and build a report from their survey photos. Instead, they switch between a file manager, a photo editor, Word, and a PDF converter, making the process unnecessarily long and error-prone.

2. **Sorting and organising large volumes** of drone images is time-consuming, increases cognitive load, and is prone to errors such as oversight, often leading to poor image selection and inconsistent report processing.

3. **Inconsistent Final Deliverable:** Without a template system, every PDF report looks different from the last and breaks, so the file that reaches the client is often misaligned or visually amateur.

4. **Scalability limitations:** As the number of surveys increases, for organisations processing operator footage, maintaining speed and quality becomes difficult without automation or additional resources, such as survey history and saved reports.

5. **Expert Knowledge Gets Lost Before It Reaches the Report:** Observations made during the flight are forgotten by the time the operator sits down to write. And without writing or agronomic training, most operators struggle to translate what they saw into clear, credible, professional language.

Product Vision and Goals

1. Streamline Post-Flight Reporting

   Vision: Revolutionise post-flight drone reporting by reducing hours of manual documentation to minutes through automation, delivering clear, structured, client-ready reports instantly.

   Goal: Automate 70–90% of the reporting workflow (photo annotation, description writing, structuring, formatting) to enable same-day reporting and replace labour-intensive manual field inspections.

2. Elevate Data Interpretation

   Vision: Use AI to extract insights from raw photos and operator notes, turning complex field observations into clear, intelligent, decision-oriented interpretations for agriculture and land management.

   Goal: Identify issues early (crop health problems, soil condition, crop stress, irrigation needs) and guide better farming and survey decisions through meaningful AI-generated insights.

3. Improve Client Communication

Vision: Deliver reports that are visually clear, professionally structured, and easy for clients to understand — strengthening user credibility and client trust.

Goal: Provide polished, concise reports that help clients quickly assess crop health, land issues, boundaries, erosion, and structural conditions without technical complexity.

4. Support Multi-Industry Use Cases & Precision Agriculture

Vision: Scale across industries from agriculture to construction and environmental surveys with accurate, data-driven insights that make farming smarter and more precise.

Goal: Enable users to switch between multiple templates (agriculture, construction, environmental) and access detailed insights into soil conditions, crop stress, and irrigation needs.

5. Centralise Client, Farm & Survey Management

Vision: Offer a unified, scalable reporting system that supports solo operators, small businesses, large agricultural firms, and farm owners by keeping all survey data organised and accessible.

Goal: Provide a single source of truth for archived reports, past surveys, farm records, and client profiles — improving long-term engagement and repeat project use.

6. Increase Accessibility for Farmers & Operators

Vision: Make advanced drone and AI technology easy for everyone, including non-technical farmers, with a simple and intuitive reporting workflow.

Goal: Ensure the platform is user-friendly enough that first-time drone operators and traditional farmers can generate reports without complexity or specialist knowledge.

7. Save Time and Reduce Costs

Vision: Make farm and land monitoring faster, cheaper, and more efficient by minimising manual labour and reducing resource waste.

Goal: Accelerate field monitoring, replace repetitive manual inspections with automated drone surveys, and reduce operational costs for both operators and farmers.

8. Promote Sustainable Agriculture

Vision: Support environmentally responsible farming by delivering insights that help users manage resources more efficiently.

Goal: Help farmers optimise water, fertiliser, and pesticide use to reduce environmental impact and improve long-term sustainability.

Target Users and Use Cases

Primary Target Users:

- **Drone Operators & Surveyors:** Professionals conducting agricultural, construction, or environmental surveys using drones. They capture large volumes of images and need to quickly convert them into structured, client-ready reports without spending hours on manual documentation.

- **Agricultural Consultants & Agronomists:**
  Experts who analyse crop conditions and provide recommendations to farmers or agribusinesses. They require clear, well-structured reports with visual evidence (images) and actionable insights.

- **Construction & Infrastructure Inspectors:**
  Individuals responsible for monitoring construction sites, roads, pipelines, or buildings. They need to document site conditions, damage, or progress in a professional, organised report format.

- **Environmental & Land Survey Analysts:**

Professionals assessing land use, erosion, deforestation, or environmental impact. They rely on detailed reports with categorised findings and supporting imagery for compliance and decision-making.

- **Small Drone Service Businesses & Agencies:** Companies offering drone survey services to multiple clients. They need an efficient system to manage multiple projects, generate reports quickly, and maintain a history of client deliverables.

Key Use Cases:

- **Use Case 1** – **Fast Post-Flight Report Generation:** A drone operator completes a farmland survey, capturing over 150 images. Instead of manually writing a report, they upload the images into FieldSpec, tag them (e.g., crop health, irrigation issues), and add brief notes. The platform automatically generates a structured report with captions, findings, and recommendations, saving hours of work.

- **Use Case 2** – **Agricultural Crop Health Reporting:** An agronomist conducts a drone survey for a large farm. After uploading and tagging images related to crop stress and pest issues, FieldSpec generates detailed descriptions and suggestions (e.g., irrigation adjustments or pest control measures). The agronomist reviews and edits the report before sending it to the client.

- **Use Case 3** – **Infrastructure Damage Inspection:** A construction inspector surveys a site and captures images of structural cracks and erosion. Using FieldSpec, they categorise the images (e.g., “infrastructure damage”) and add notes. The system generates a professional report that highlights key issues, making it easy to present the findings to stakeholders or clients.

- **Use Case 4** – **Environmental Impact Assessment:** An environmental analyst conducts a land survey to assess erosion and vegetation loss. After uploading and tagging images, FieldSpec produces a report with organised sections, annotated visuals, and summary insights that can be used for compliance or regulatory submissions.

- **Use Case 5** – **Multi-Client Report Management:** A drone service agency handles surveys for multiple clients weekly. Using FieldSpec, they generate and store reports

under each client profile, track past surveys, and quickly retrieve or update reports when needed. This improves organisational efficiency and client service.

Key Features (MVP)

The Minimum Viable Product will focus on features that enable drone photos to be turned into structured reports quickly using AI.

1. Photo Upload & Organisation

- Users can upload multiple survey photos (drag-and-drop or file picker)
- Basic tagging of images by category (e.g., crop health, erosion, damage)
- Filter for categories, e.g., filter for images with bad crop health

2. AI Report Generation

- Generate structured report sections (captions, findings, summary) from user notes and tagged images
- Auto-create descriptions for each image

3. Simple Report Template

- One default report template (no multiple industries yet)
- Predefined sections: Introduction, Findings, Recommendations, Image Annotations

4. Editable Report Preview

- Users can review and edit AI-generated content before export
- Basic text editing (no advanced formatting needed)

5. Option for file export

- Export the final report as a downloadable PDF, CSV, Excel, or other format. Clean, client-ready layout

6. Basic Project Saving

- Save reports locally or in-app
- Resume editing previously created reports

7. Insight Trace

- Shows source links between images, notes, and insights. Also shows a small label “AI-Generated”. So the user clicks on “Insight”, and it shows them the images that the report was generated from. We will also add a confidence score to the document to indicate the reliability of the AI-generated report. These build confidence through transparency

8. Map-Based Visualisation

- Display uploaded survey images as markers on a map using their location data
- Use colour-coded markers to represent different categories (e.g., crop health, erosion, damage)
- Allow users to click on markers to view image previews and notes
- Enable manual adjustment of marker positions for accuracy
- Provide filtering options to view specific categories on the map
- Generate a map snapshot to be included in the final report

9. Project Dashboard & History

User can: View and manage all past survey projects.

Dashboard includes:

- List of all projects (card or table view)
- For each project: Name, Date, Industry, Photo Count, Last Edited
- Status badges: Draft/ Report Generated /Exported
- Quick actions: Continue Editing /Generate New Report |/Export PDF / Duplicate Project
- Search & Filter:
- Search by project name or client
- Filter by industry or date range

10. App Settings (Minimal)

MVP may support a single user per account (one login)

- Update basic profile info (name, email, or company name for reports)

Future Features & Roadmap

As FieldSpec grows, we look to offer enhancements that deepen value after the product is proven in the market.

Smarter Workflows (Phase 2)

- Multi-user team accounts with role-based access (Admin, Pilot, Report Writer)
- Survey history comparison to compare two reports for the same site side by side
- Custom report templates so operators can build and save their own layouts
- Mobile-optimised field note capture during or right after a flight

Business Operations (Phase 3)

- Client portal: clients get a link to view and download their own reports
- Automated re-survey reminders (e.g. "Client X's last survey was 90 days ago")
- The reporting analytics dashboard tracks surveys completed, findings, trends, and client activity

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Platform & Integrations (Phase 4)

- Integration with drone flight apps (DJI, DroneDeploy) to auto-import photos post-flight
- Integration with farm management platforms (John Deere Operations Centre, Trimble)
- Offline mode for operators in remote areas with no internet during report building

Road map

Q1 (Month 1-3)

MVP: BUILD AND LAUNCH
M1: Discovery and Research
M2: Build
M3: Beta testing

Q2 (Month 4–6)

PHASE 2: SMARTER WORKFLOWS
M4: Team accounts
M5: Survey comparison
M6: Custom templates

Q3 (Month 7-9)

PHASE 3: BUSINESS OPERATIONS
M7: Client portal
M8: Re-survey alerts
M9: Analytics dash

Q4 (Month10–12)

PHASE 4: PLATFORM & INTEGRATIONS
M10: Drone flight apps integration
M11: Integration with farm management platforms
M12: Offline mode

Free vs Paid Tier

FieldSpec would adopt a Freemium model, with a starter plan to encourage new users to try FieldSpec's features and a more value-added plan for power users.

Starter Tier-(Launch and Starter plan)

Designed for individuals or small farm operators to try out FieldSpec

Price- **Free**

Available features

- **5 projects/month**
- **50 images per project**
- **Basic templates:** Access to a single or very limited report structure.
- Fixed sections:
    - Introduction
    - Findings
    - Recommendations
    - Image annotations
- **Upload + tagging:** Users upload images and categorize them(Drag-and-drop upload, Manual tagging)
- **AI captions & findings:** AI generates basic insights from images.**(**Per-image captions, Simple findings[no deep cross-image reasoning])
- **PDF export:** Users can download a clean, shareable report with a structured layout, images, captions, findings, and minimal styling.

Professional Tier-(Core revenue plan)

Designed for active drone operators, small businesses, agronomist

Price- $20 / month

Available feature

- **35 projects/month**
- **150 images per project**
- **Advanced AI report generation:** A more sophisticated AI system that produces context-aware, structured, and reliable reports from images + user input.
- **Multiple templates (agriculture, construction):** Predefined report formats tailored to different industries.
- **Editable report builder:** A structured interface for modifying AI-generated reports before export.
- **Map display power:** A data visualisation + reporting engine tied to geolocation,mapsnapshot in the reports, marker editing and the user can filter by

Category (crop health, erosion, damage), Severity (if available later), Tagged vs untagged images.

- **Client management system:** A centralised system to manage all client-related information and link it to projects.
- Client profiles (name, company, contact details)
- Project association:
    - History tracking
    - Past reports per client
    - Each project is linked to a client.

Business/Team Tier-(Core revenue plan)

Designed for Drone companies, Agencies, and medium teams

Price- $299 / month

Available feature:

- **Unlimited projects (fair usage):** Users can create unlimited projects, but within reasonable system limits.
    - No hard cap on the number of projects
    - Backend enforces **fair usage policy**
- **High image limits:** Higher cap on number of images per project (e.g., 800–2000 images/project) and Optimised upload + processing pipeline
- **Team collaboration:** Multiple user accounts per workspace, Shared access to projects
    - Basic roles: Admin (full access), Editor (edit reports), Viewer (read-only)
- **Shared client database:** Centralised storage of client information accessible by team members.
- **Full Map power:** A data visualisation + reporting engine tied to geolocation, Map snapshot in reports, Marker editing and more advanced filtering

- **Priority AI processing:** Faster AI report generation compared to lower tiers**.** Jobs from Business users are processed first using the Queue system (e.g., BullMQ) or Priority levels
- **Branding (custom logos on reports):** Ability to customise report appearance with company identity.
    - Upload the company logo
    - Display logo on report header/cover page
    - Optional company name/footer

Enterprise Tier-(Custom pricing)

Designed for large agriculture firms, construction companies, government / NGOs

Price- **Custom ($500+/month or contract-based)**

Available Features:

- **API access:** Programmatic access to FieldSpec functionality. Allows integration into existing pipelines
- **Integrations (e.g. DroneDeploy):** Potential Integration Ability to connect FieldSpec with external systems.
- **Dedicated support:** Priority access to human support with guaranteed response times, a dedicated Account manager, and 24/7 support
- **SLA guarantees:** A **Service Level Agreement** guaranteeing system performance(AI processing time guarantee, e.g., “Report generated within 2 minutes for ≤300 images”, Uptime guarantee (e.g., 99.5%), Priority queue access)
- **Custom templates:** Client-specific report structures and branding(Custom sections, Custom ordering of sections, Branding[logo, Colours, company details]) and Reusable templates per account.

Detailed User Flows

Below are the detailed step-by-step user flows for key activities in FiledSpec

Flow 1: User Onboarding

Goal: Create an account and start the first report in under 3 minutes.

- Sign Up – Email + password OR Google OAuth.
- **Onboarding:** Basic Profile: Enter name + company name (optional, appears on report cover)
- Dashboard: User lands on empty project dashboard with one button: "+ New Report"

Flow 2: Post-Flight – Upload & Organise Images

Goal: Upload 100+ images, tag by category, and add brief notes.

- Start New Report – Click "+ New Report" → Enter project name (e.g., "Smith Farm – North Field")
- Upload Images – Drag & drop or file picker (JPEG/PNG/TIFF). Progress bar for large uploads.
- Auto-Sorting (basic) – System suggests categories based on EXIF GPS/time grouping. User confirms or re-assigns.
- Tagging & Notes – Per image: select category from list (crop health, erosion, damage, irrigation, general), add short notes. Bulk tag multiple images.
- Thumbnail Grid – Review all images, filter by category, and delete/replace if needed.

Flow 3: Map Review

Goal: Validate geolocation and visualise survey coverage.

- View Map – After upload, the user sees a map with image markers colour-coded by category.
- Inspect Marker – Click marker → image preview + tags + notes pop up.
- Adjust Position – Drag the marker to correct GPS inaccuracy.
- Filter Map – Toggle categories on/off (e.g., show only "erosion" markers).
- Capture Map – Click "Save Map View" → snapshot will appear in the final report.

Flow 4: Report Builder & Insight Trace

Goal: Generate structured reports with transparent AI insights.

1. Generate Report – Click "Generate Report" → system creates:
- Cover page (project name, date, user's company name)
- Introduction (auto-generated from project name + date)
- Section per category (e.g., "Crop Health Issues")
- Each image has a caption (pre-filled from the user's note)
- Summary of findings (AI-generated from tags + notes)
- Recommendations (simple rule-based: e.g., "irrigation issue" → check water distribution)

2. Review with Insight Trace – For each AI insight:
- Shows confidence score (e.g., "92% confidence")
- "View Source" button → highlights images/notes that generated this insight
- Users can Accept, Edit, or Reject each insight

3. Manual Edits – Reorder sections, edit any text, add/remove images, adjust captions.
4. Preview Report – PDF-like preview with clean layout (no branding/custom templates).
5. Save as Draft – Auto-saves to cloud.

Flow 5: Export & Archive

Goal: Deliver PDF and save for future reference.

1. Final Review – System checks for untagged images or blank fields → shows warnings.
2. Export Options – Download as PDF (standard), or generate a shareable link.
3. Archive – Report auto-saved to dashboard with metadata (project name, date, photo count).

4. Return to Dashboard – User sees updated project list.

Flow 6: Project Dashboard & History

Goal: Manage all past reports.

1. Dashboard View – Card/table list showing:
- Project name, Date, Photo Count, Last Edited
- Status badge: Draft / Report Generated / Exported

2. Quick Actions – Continue Editing / Generate New Report / Export PDF / Duplicate Project
3. Search & Filter – Search by project name, filter by date range.

Technical Requirements

FrontEnd Architecture

To maintain a high-performance, browser-native experience, FieldSpec will utilise a modern reactive framework

A. React / Next.js

A web client (Hybrid SPA: Single Page Application) built with a modern framework (e.g.,Next.js App Router). This will handle the user interface, display of data, and user interactions, also for routing the system, to provide backend API(uploads & generate reports), performance optimisation (image, code splitting e.t.c)

B. Tailwind CSS (fast UI iteration)

To help build image grids, report layouts, tags, editor blocks, dashboards and whatever ui components, especially those that keep changing constantly, for instant styling using utility classes, a consistent spacing system, faster iteration speed

C. State management: Zustand or Redux Toolkit

Zustand is Lightweight, very fast setup, Great for complex UI state (your case) and would provide minimal boilerplate

Best for: image selection, report editing state, UI interactions, AI status tracking

D. Routing:

Client-side routing to manage different survey views (Gallery, Annotation Mode, Report Preview) without server requests.

Key Requirements

- Drag-and-drop file handling (use `react-dropzone`)
- Real-time UI updates (no reloads)
- Optimistic UI for tagging + notes(e.g., crop health, erosion, damage)
- Dashboard for viewing past surveys and reports
- Dynamic report builder with editable sections
- Virtualised image grid (for hundreds of photos)

Backend Architecture (Control Layer)

FieldSpec will use Node.js, Next.js API routes, or Express to handle sensitive, structured workflows such as file uploads, AI generation, report creation, and client data storage. The backend is required to securely manage API keys for DeepSeek, process and queue long-running AI tasks (so the UI doesn't freeze), validate incoming and outgoing data using Zod to prevent broken or inconsistent reports, handle secure image uploads (e.g., signed URLs), and persist all project data reliably in a database.

Stack

- Node.js (Next.js API routes or Express)
- Type-safe APIs (Zod validation)

A. API Requirements

Core Endpoints

- `/upload/initiate` → returns signed upload URL
- `/images/metadata` → store tags + notes

- `/ai/generate` → queue AI jobs
- `/report/export`
- `/ai/status/:jobId`
- All endpoints must be **idempotent** (safe retries)

B. Queue System (MANDATORY)

- **Redis + BullMQ**

Turn slow AI and image processing into controlled background tasks so the app stays fast, stable, and scalable even with hundreds of images per report.

Why is this critical

- AI generation is asynchronous
- Prevents:
  - API blocking
  - Rate limit crashes
  - Timeouts

Queue Jobs

- Image caption generation
- Section summary generation
- Full report synthesis

All endpoints must be **idempotent** (safe retries)

AI-LAYERED(Controlled IntelligenceSystem)

Using: DeepSeek

- API Gateway: A thin, secure backend proxy (Node.js/Edge Functions) to communicate with the DeepSeek-V3 API.

The system must ensure:

- AI outputs are predictable, structured, and editable
- UI remains fast with 200–500 images
- Reports require near-zero polishing before client delivery.

A. Input Structure (Strict Schema)

AI must NOT be free-form. It must be boxed into a deterministic system.

Every AI request must follow:

- "category": "crop_health",
- "user_note": "Yellowing leaves in section B",
- "context": "Maize farm, 5 acres"

Requirements

- No raw text prompts from UI
- All inputs are normalised before sending
- Categories must be predefined enums

B. Prompt Engineering Requirements

You must enforce:

- Role definition:
  "You are an agricultural survey analyst..."
- Output format (STRICT JSON):

★ "caption": "string",
★ "finding": "string",
★ "recommendation": "string"

C. AI Validation Layer

Before saving AI output:
Context Window Management: Optimisation of token usage by summarising field notes locally before sending them to the LLM.

➢ JSON schema validation (Zod)
➢ Max tokens per request
➢ Monthly usage tracking
➢ Reject invalid responses
➢ Retry with stricter prompt OR flag for manual edit

To prevent hallucination + formatting inconsistency

D. Human-in-the-Loop Editing

- Every AI output must be editable
- Track edits vs original AI output

E . Batch Processing Strategy

- Process images in chunks (10–20 per batch)
- Parallel but rate-limited
- Queue-based execution (Redis + BullMQ)

  To prevent system failure and API failure

F. Fallback strategy

- “Generate empty structured template”
- Allow manual completion
-

Image Storage & Delivery

Stack

- AWS S3 (or Cloudinary)

To handle large volumes of images, they act as external, scalable storage systems optimised for media and storing or serving them directly from your backend would be slow, expensive, and unreliable.

Requirements

- Signed URL uploads (secure, direct-to-cloud)
- Store:
  - Original image
  - Thumbnail
- CDN delivery (CloudFront, images load quickly even with a weaker network)
- Thumbnail-first loading
- Lazy loading (Intersection Observer)
- Virtualised grid (react-window)

Storing or serving them directly from your backend would be slow, expensive, and unreliable.

Database Design (Structured + Scalable)

Stack

- PostgreSQL

To store system data: Clients, Survey projects, Hundreds of images per project, Tags + notes per image, AI-generated outputs (caption, findings, recommendations), and final reports.

- AWS S3 (or Cloudinary)

To handle **large volumes of images**

- Redis + BullMQ

Act as task manager for job queues, retries and background workers (AI tasks are asynchronous)

- CACHing Layer(optional)

  To cache frequent queries and store temporary states

Core Tables

Users

- ID, email, auth_provider

Clients

- ID, name, contact_info

Projects (Survey Sessions)

- ID, client_id, location, date

Images

- ID, project_id, url, category, notes

AI jobs

- ★ ID, project_id, status, progress

AI_Outputs

- ★ image_id, caption, finding, recommendation

Reports

- ID, project_id, template_type, status

Report Engine (Core Product)

- Dynamic Layout Engine: A system to convert the SPA's "Preview" state into a paginated PDF format, ensuring annotations are correctly baked into the exported images.

A. Structured Report Engine

Reports MUST follow a strict hierarchy

```
 "title": "Survey Report",
 "sections":
   "Executive Summary",
   "Findings",
   "Image Analysis",
   "Recommendations"
```

B. Template System

Templates must define:

- Section order
- Tone (formal, technical)
- Industry-specific language

Supported Templates

- Agriculture
- Construction
- Environmental

Ensures: professional consistency

C. Content Standardization

AI Output Rules

- No slang
- No uncertainty language ("maybe", "possibly")
- Use domain-specific terminology

D. Rendering Engine

- HTML-based rendering (React components)
- Dynamic section injection

E. PDF Export Engine

- html2pdf.js

Requirements

- Print-specific CSS
- Page break rules
- Asset preloading before export

Prevents: broken PDFs

Security Requirements

A. JWT(JSON Web Tokens) authentication

JWT (JSON Web Token) is used to verify that the request is coming from a logged-in, valid user, ensuring every action in the system is tied to a verified user identity

B. Signed upload URLs

A **signed URL** is a temporary, secure link generated by the backend that allows users to upload files **directly** to storage (like AWS S3). This ensures that users can upload large files safely and efficiently without exposing the system to crashes, slow performance and server overload

C. Rate limiting (AI endpoints)

How many requests can a user send in a given time? To prevent abuse of AI calls because it costs money and consumes resources & to stabilise performance(e.g., 10 requests/sec)

D. Input sanitisation (prevent injection into prompts)

cleaning and validating user input before using it. Ensures users' input cannot break the AI, your data, or your reports

a. Removes harmful characters/scripts
b. enforces expected formats
c. filters dangerous instructions before AI

Metrics of Success

To measure the effectiveness and adoption of FieldSpec, we will track metrics across the following core outcomes:

1. Efficiency & Time Savings:

Target: Reduce report creation time from the manual 2–6 hours down to 10–15 minutes, with at least 60% time saved per report.

Metrics:

- Average report creation time (upload → final PDF export)
- Time saved per report compared to manual workflow

- Photos processed per session (>100 images without performance issues)
- Average upload-to-tagging speed (<10 seconds per photo)
- Onboarding-to-first-report time (<20 minutes)

Explained:
This metric evaluates how effectively FieldSpec reduces the heavy manual workload of report creation. It measures how quickly users move from photo upload to final PDF export, how many photos can be handled per session, and the speed of tagging and onboarding. All indicators collectively show how much operational efficiency the product provides to drone operators.

2. AI Output Quality:

Target: Achieve high-quality AI-generated content requiring less than 30% edits, with overall user satisfaction ≥ 4/5.

Metrics:

- User edit rate on AI-generated text (<30%)
- AI response time (<3–8 seconds per section/photo)
- Report completion success rate (>85%)
- User satisfaction score for report quality (≥4.0/5)

Explained:
This metric focuses on how effectively the AI generates captions, summaries, and findings. Low edit rates, fast generation times, and high satisfaction scores indicate that the AI is producing reliable, high-quality content that reduces the manual editing burden for users.

3. User Adoption & Retention:

Target: Drive consistent usage with strong repeat behaviour and growth: WAU increasing 10–15% month-over-month and a 30-day retention rate >40%.

Metrics:

- Weekly/Monthly Active Users (WAU/MAU)
- Growth in WAU (+10–15% MoM)
- Reports generated per user per week (2–3+ reports)
- 30-day retention rate (>40%)
- Repeat usage rate (>60% return within 7 days)

Explained:
Adoption and retention measure whether users find FieldSpec valuable enough to return and generate multiple reports. Strong weekly activity, consistent MoM growth, and frequent report creation indicate that the product fits well into users' workflow and is sticky enough to keep them active.

4. Export & Delivery Success:

Target: Ensure reliable export performance, with PDF export working 98% of the time and under 10 seconds.

Metrics:

- PDF export success rate (>98% without errors)
- Average export time (<10 seconds)
- Reports shared/downloaded (>90%)
- Report completion rate (started vs exported)

Explained:
Exporting is the final output of the FieldSpec workflow, making reliability essential. A high export success percentage, fast export times, and high share/download rates

signal that the system is technically stable and that users consistently finalise and deliver reports to clients.

5. Report Accuracy & Relevance:

Target: Reach 90% user satisfaction with how accurate and relevant the AI findings and summaries are.

Metrics:

- Post-edit AI change rate (%)
- Accuracy/quality feedback rating
- Client satisfaction ratings (professionalism of reports ≥85%)
- % of users who continue using default templates without major edits

Explained:
This metric evaluates whether the generated reports correctly reflect field conditions and are trusted by users and their clients. Low edit rates and strong satisfaction scores show that the findings are meaningful, relevant, and presentation-ready, reducing unnecessary rework.

6. Automation Coverage Rate:

Target: Ensure AI automates at least 80% of the reporting workflow, including narrative writing, structured summaries, and basic annotation.

Metrics:

- % of report content generated/suggested by AI
- % of photos requiring manual rewrite
- AI feature adoption (used by >80% of users)
- Processing latency for AI generation (<5–8 seconds per section)

- System reliability & uptime (>99–99.5%)

Explained:
Automation coverage measures how much of the workload the AI handles effectively. High use of AI tools, minimal manual rewriting, and consistent system reliability indicate that the platform successfully reduces user effort and scales.

Risks & Limitations

1**. AI Accuracy Risk**

AI-generated insights may be incorrect if user input is poor or ambiguous.

- Impact: Misleading reports
- Mitigation: Allow full user editing, validation, and display confidence scores

2. **Image Quality Dependency**

Low-quality drone images reduce the accuracy of analysis.

- Impact: Weak or incorrect findings
- Mitigation: Provide image quality guidelines and real-time warnings

3. **User Input Dependency**

The system relies on proper tagging, notes, and user categorisation.

- Impact: Poor outputs if users misuse or skip inputs
- Mitigation: Add smart suggestions and auto-tagging (future feature)

4. **Limited Domain Intelligence**

AI may not fully understand complex agricultural or environmental conditions.

- Impact: Oversimplified or inaccurate recommendations
- Mitigation: Position AI as an assistant, not a final decision-maker

5. **Data Privacy & Security Risks**

Sensitive land and client data are stored within the system.

- Impact: Legal issues and loss of user trust if breached
- Mitigation: Implement encryption, authentication, and secure storage

6. **Scalability Constraints**

Handling large volumes of drone images may affect system performance.

- Impact: Slow uploads and processing delays
- Mitigation: Optimize image compression and enable batch processing

7. **Large File Handling**

Drone image and video files are typically large and resource-intensive.

- Impact: Upload failures or long processing times
- Mitigation: Apply file compression and enforce file size limits

8. **User Trust in AI**

Users may doubt or question AI-generated insights.

- Impact: Low adoption and reduced reliance on the system
- Mitigation: Provide visual highlights and clear, explainable outputs

9. **Internet Dependency**

The system requires a stable internet connection to function effectively.

- Impact: Disruptions in upload, processing, and report generation
- Mitigation: Optimize data usage and introduce offline mode (future feature)

10. **MVP Limitations**

The initial version may have limited features and capabilities.

- Impact: Reduced functionality and user experience gaps
- Mitigation: Plan iterative updates to improve AI, UI, and real-time processing